

Model Analysis Report

Generated: March 04, 2026

Dataset Information

Number of timesteps: 136214

Spatial grid: 50 latitudes × 24 MLTs

Time range: 2019-11-01 00:00:00 to 2024-01-31 23:56:00

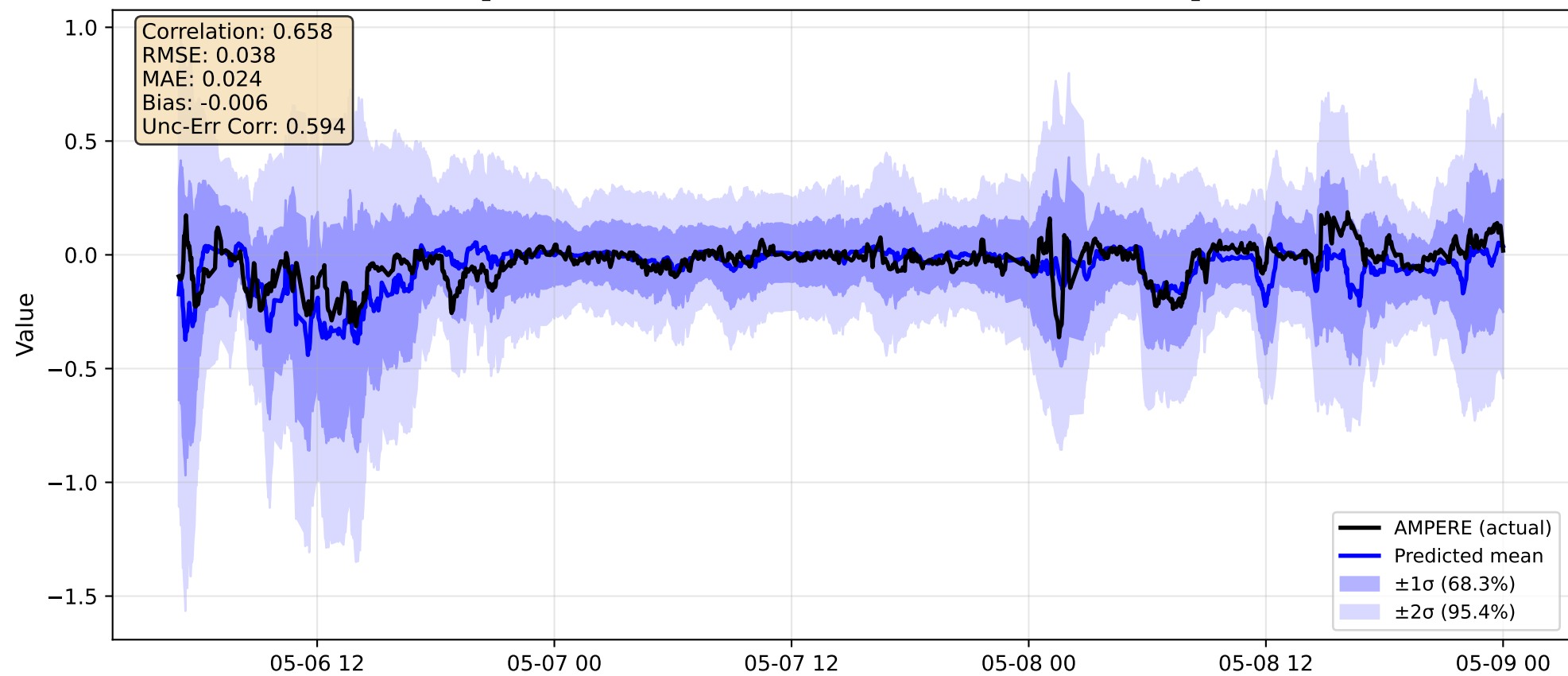
Uncertainty available: Yes

Analysis location: Lat 65-75°, MLT 10-14

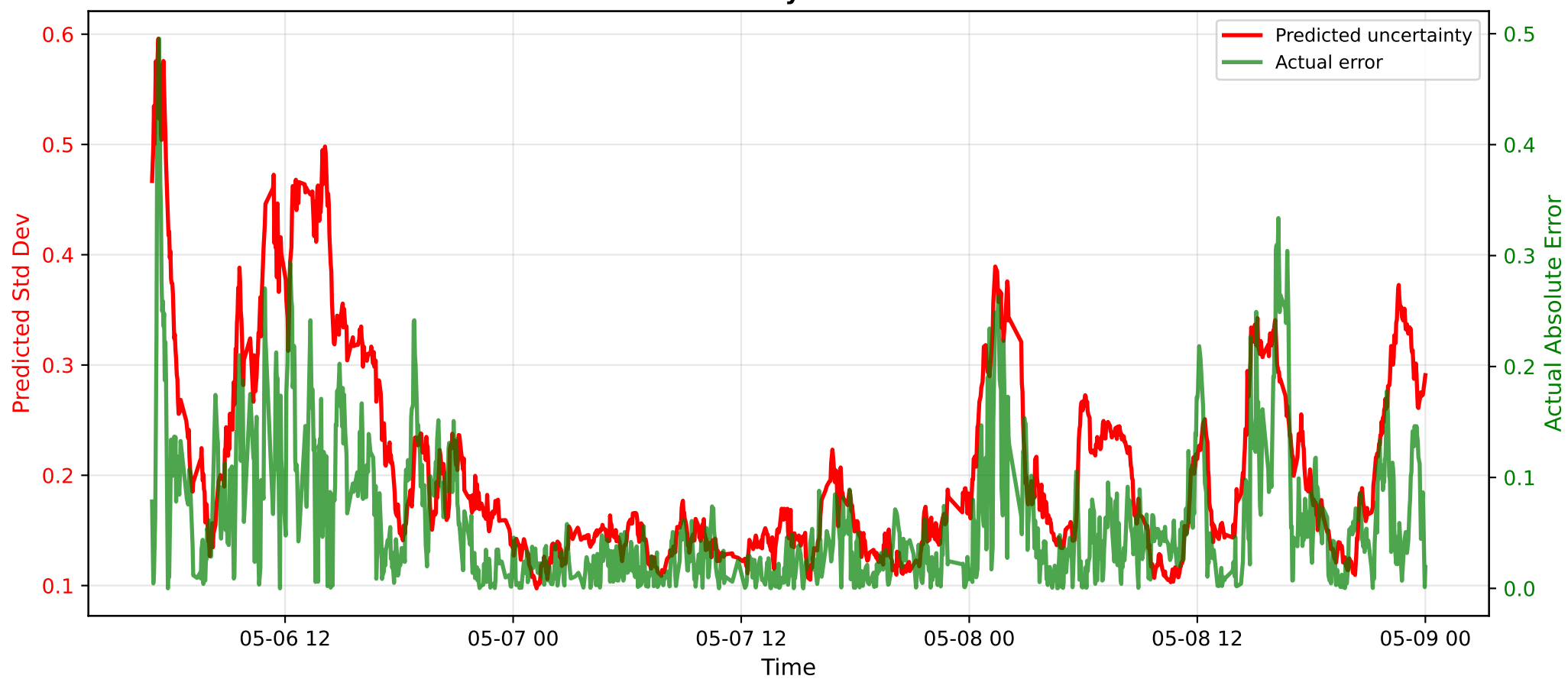
Plot time range: 2023-05-04 00:00:00 to 2023-05-09 00:00:00

*This report contains comprehensive validation metrics
for geomagnetic model predictions*

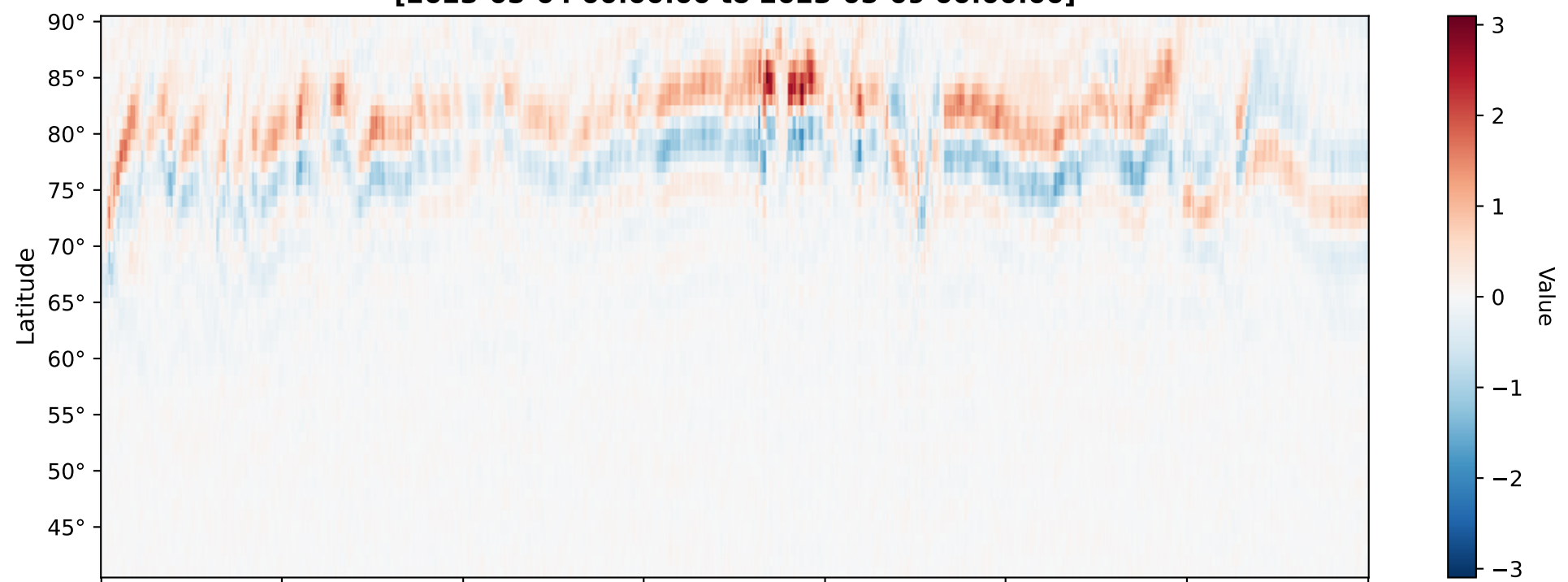
Time Series with Uncertainty (Lat 65-75°, MLT 10-14)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]



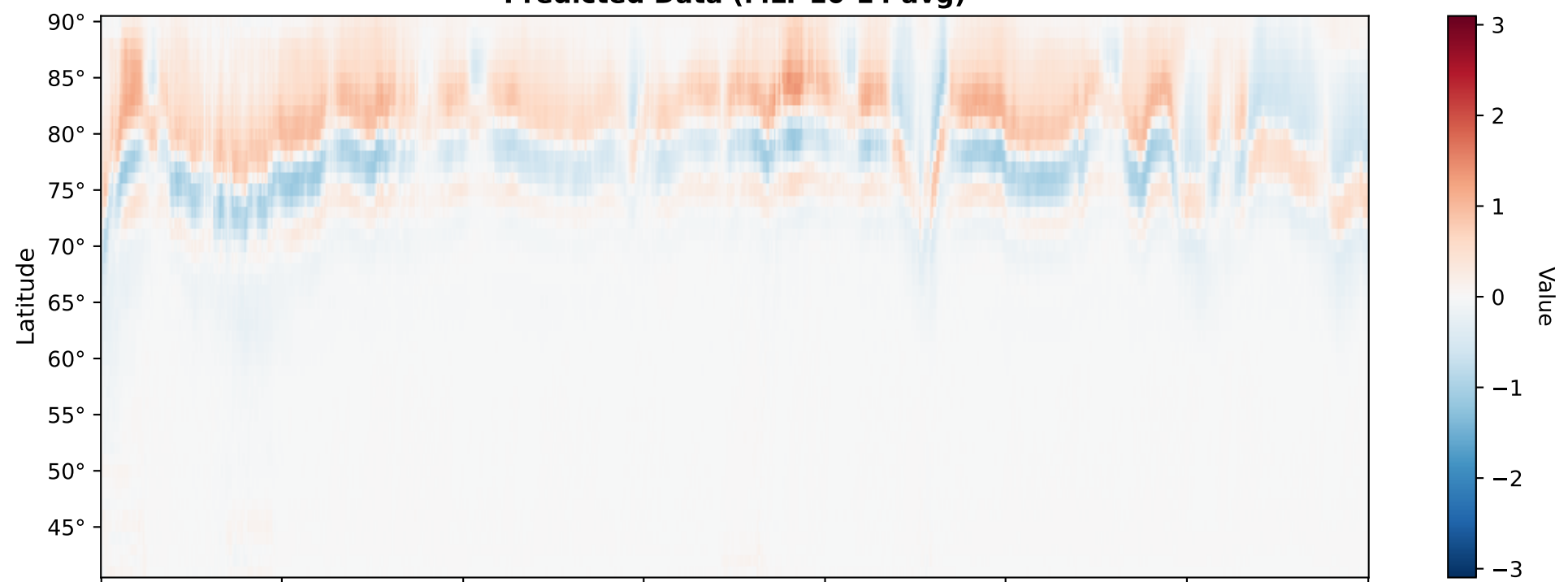
Uncertainty Evolution



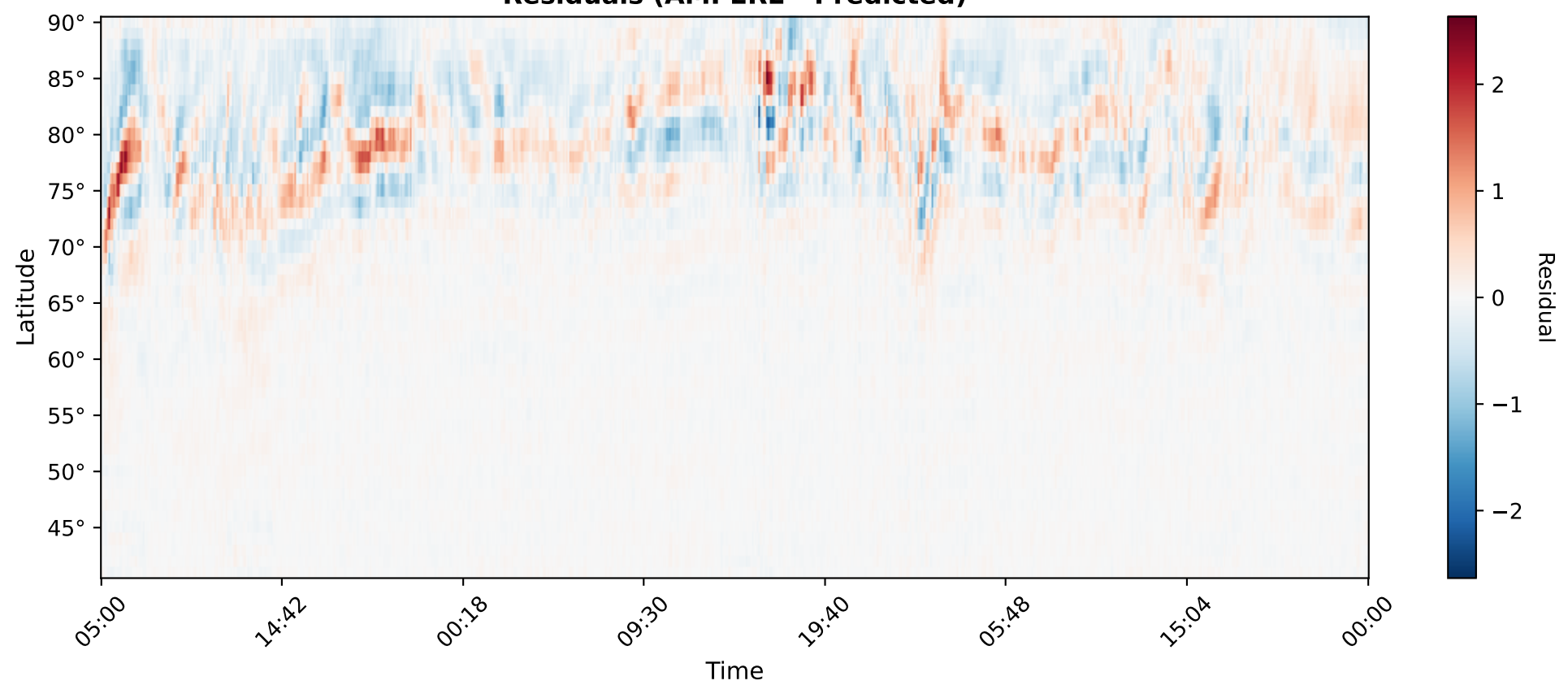
AMPERE Data (MLT 10-14 avg)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]



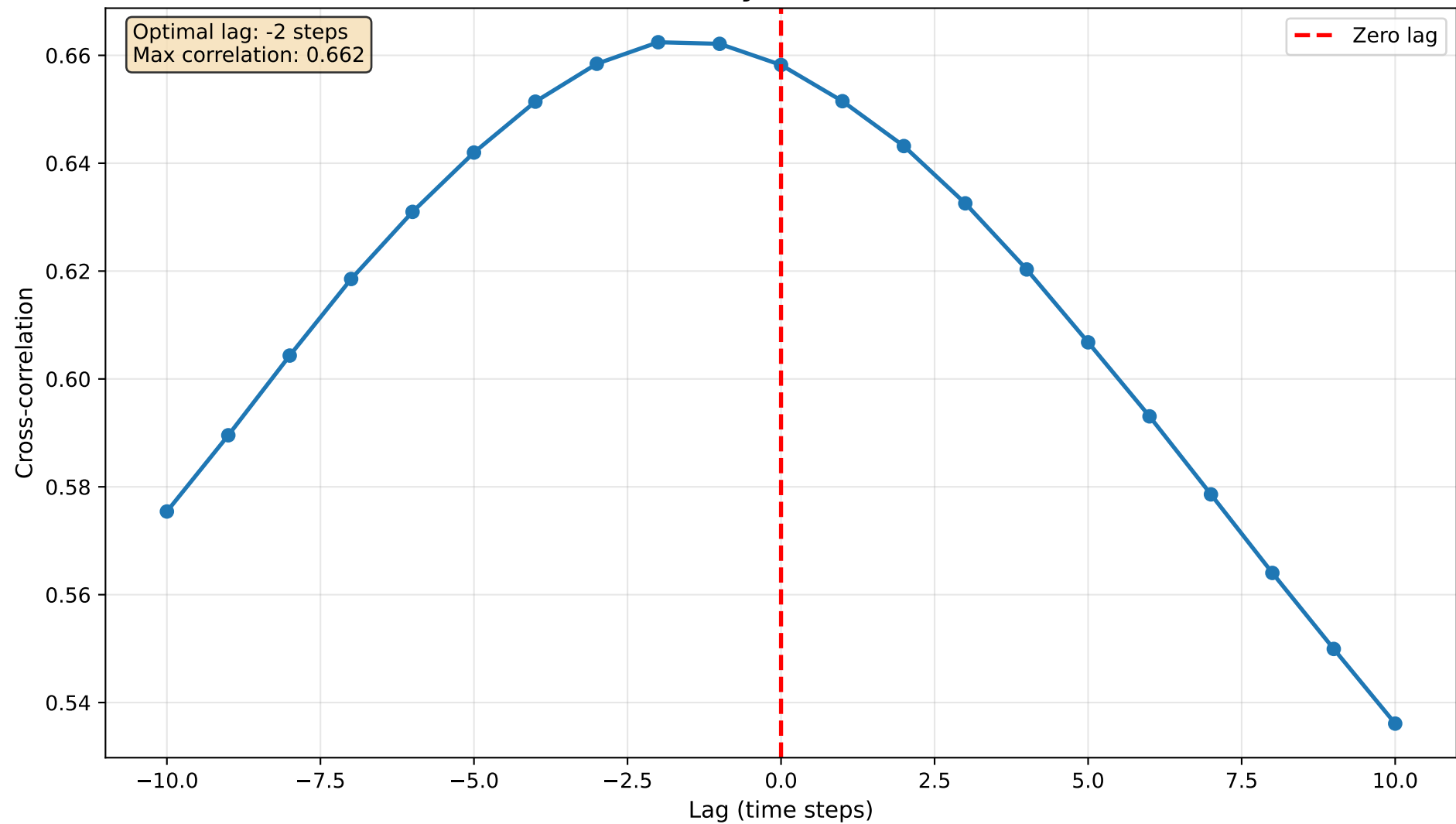
Predicted Data (MLT 10-14 avg)



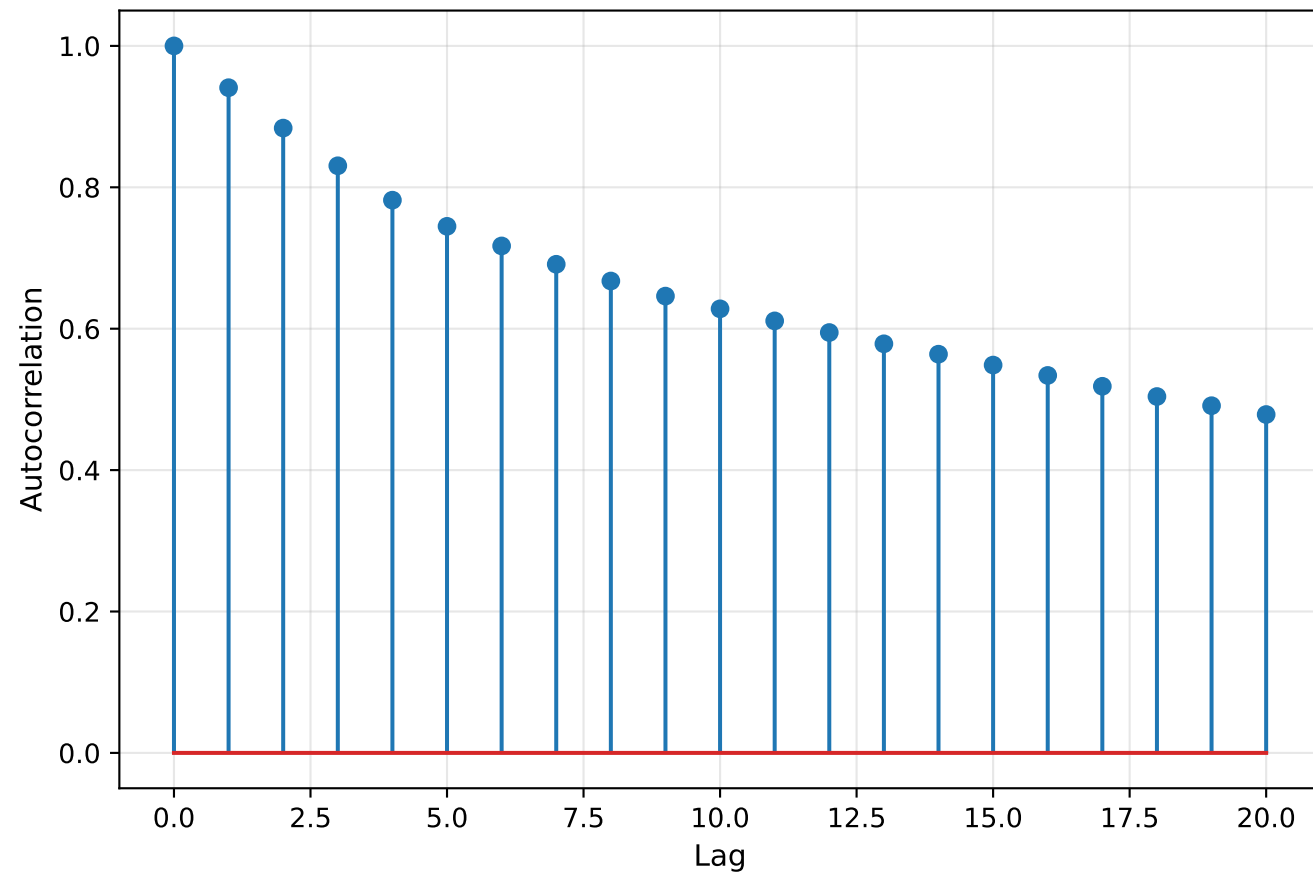
Residuals (AMPERE - Predicted)



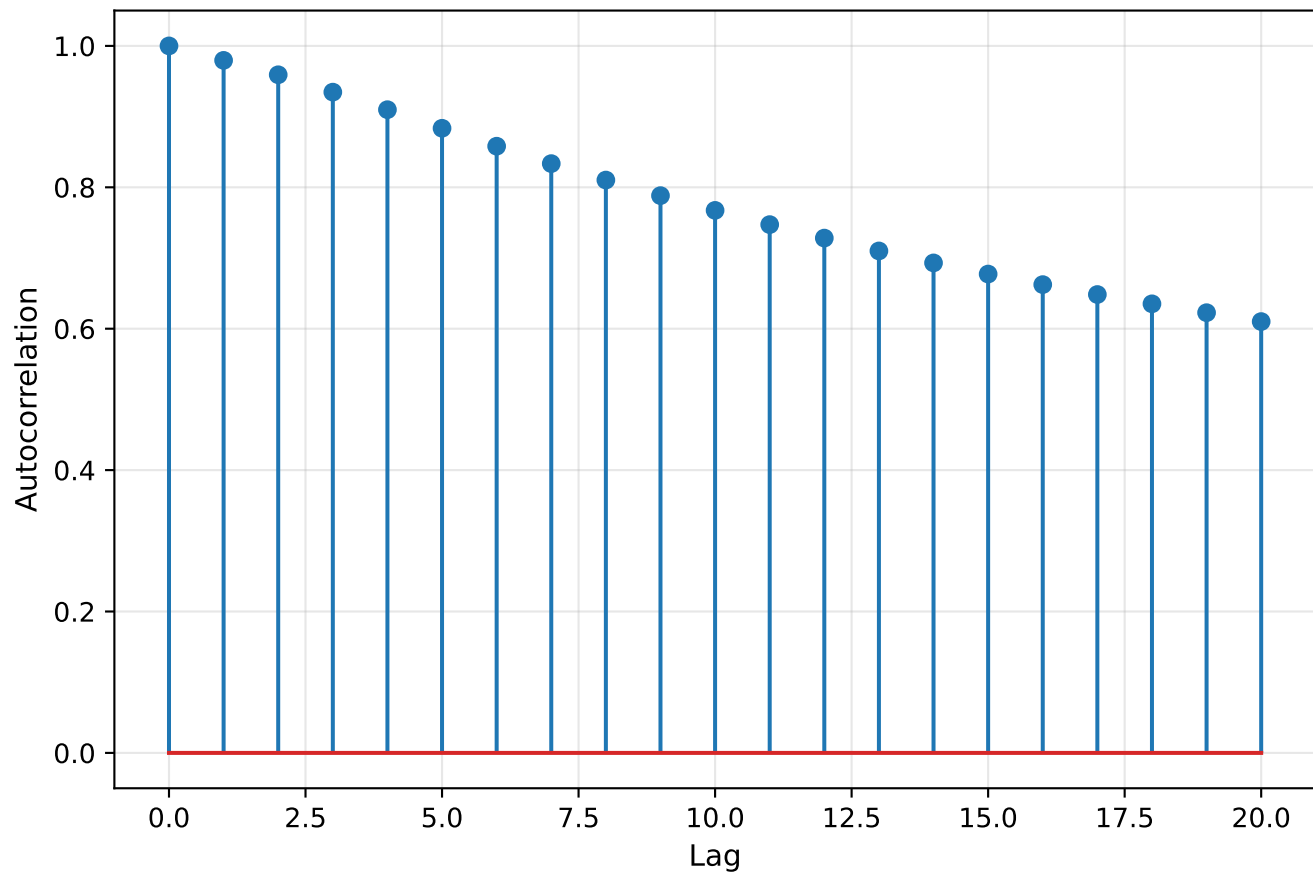
Cross-Correlation Analysis (Lat 65-75°, MLT 10-14)

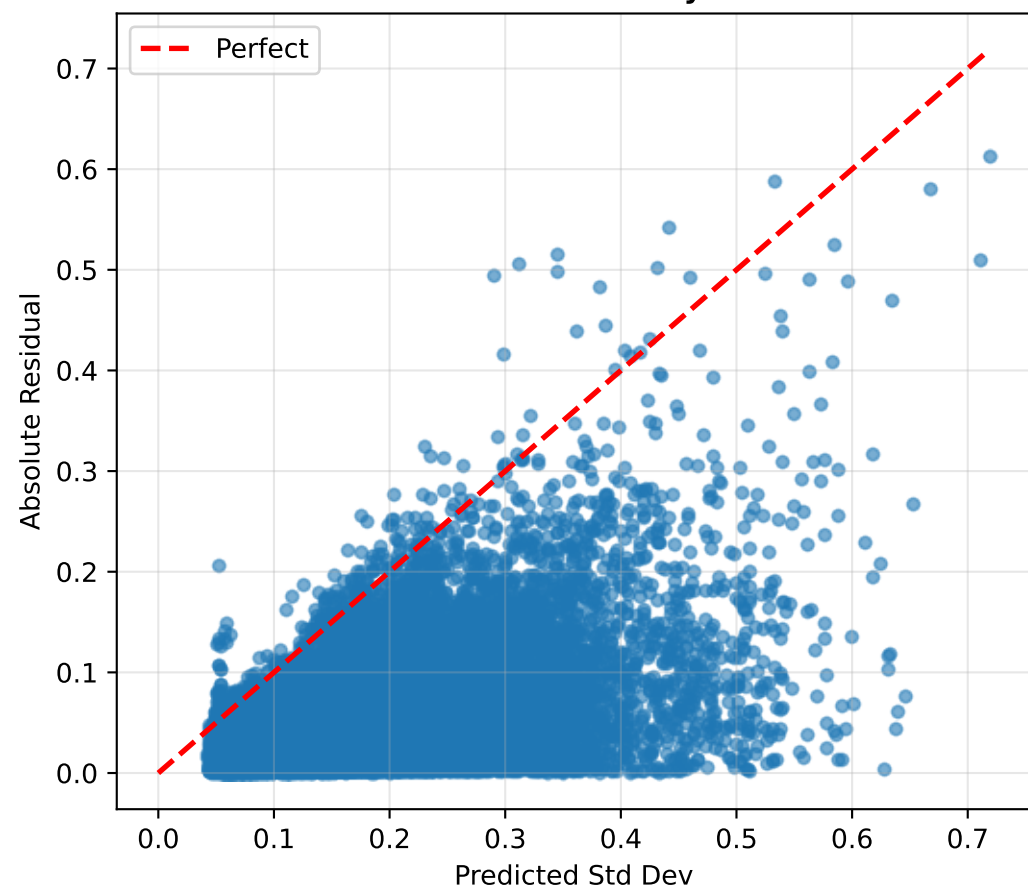
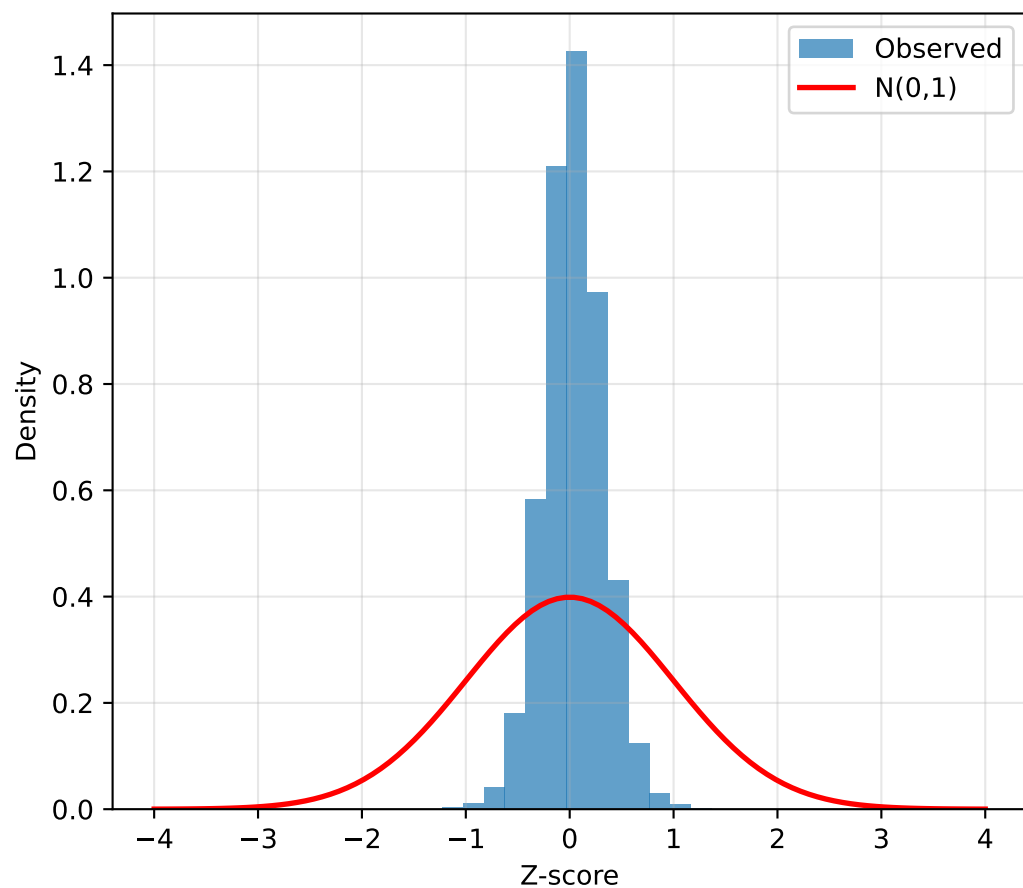
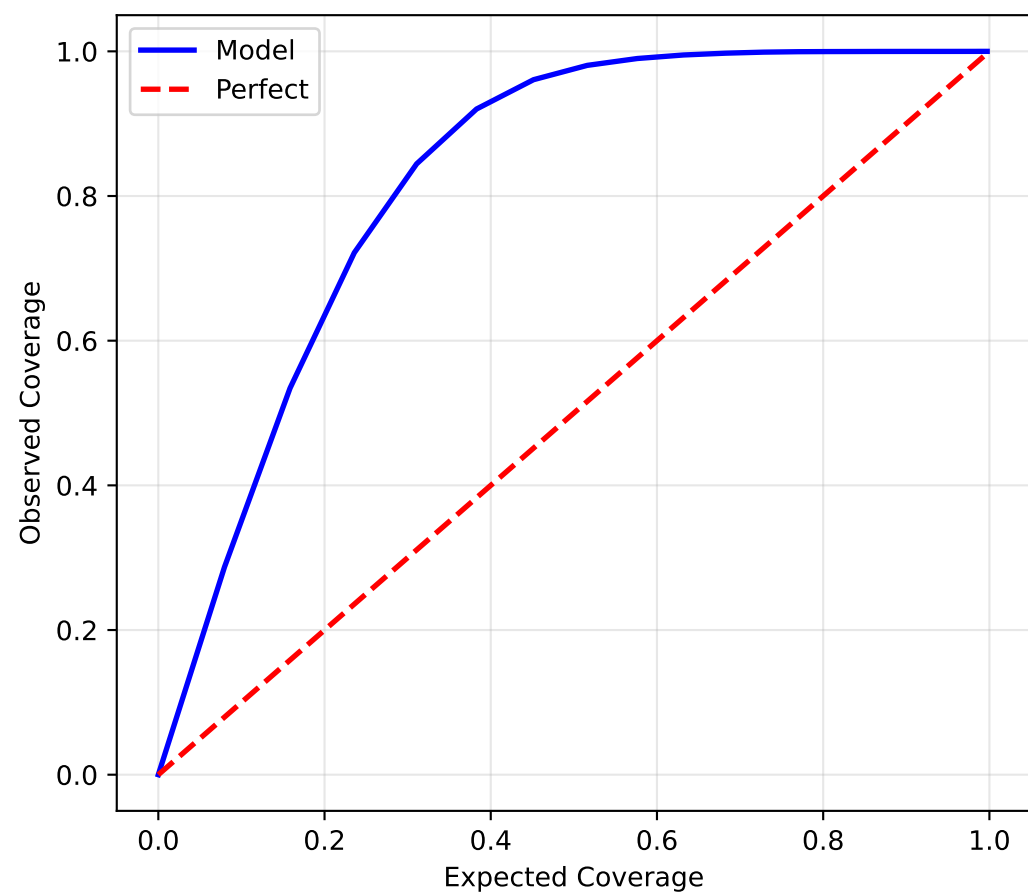
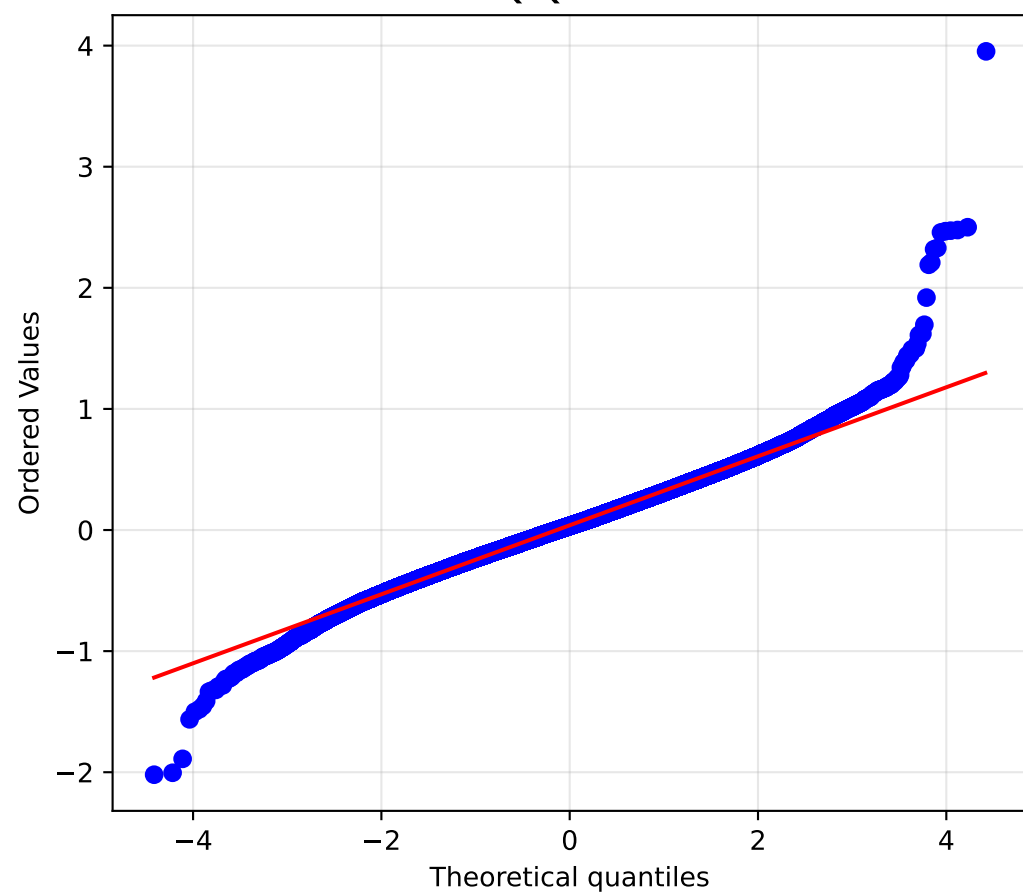


AMPERE Autocorrelation (Lat 65-75°, MLT 10-14)

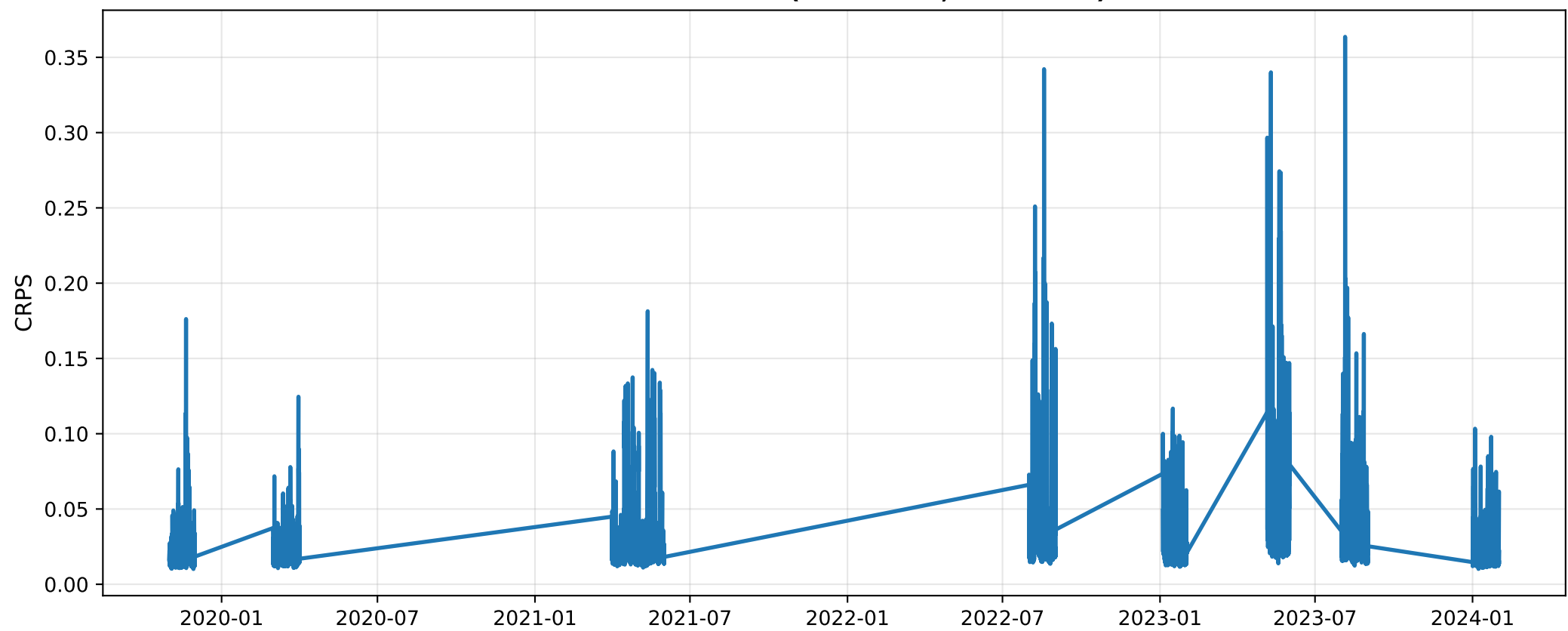


Predicted Autocorrelation (Lat 65-75°, MLT 10-14)

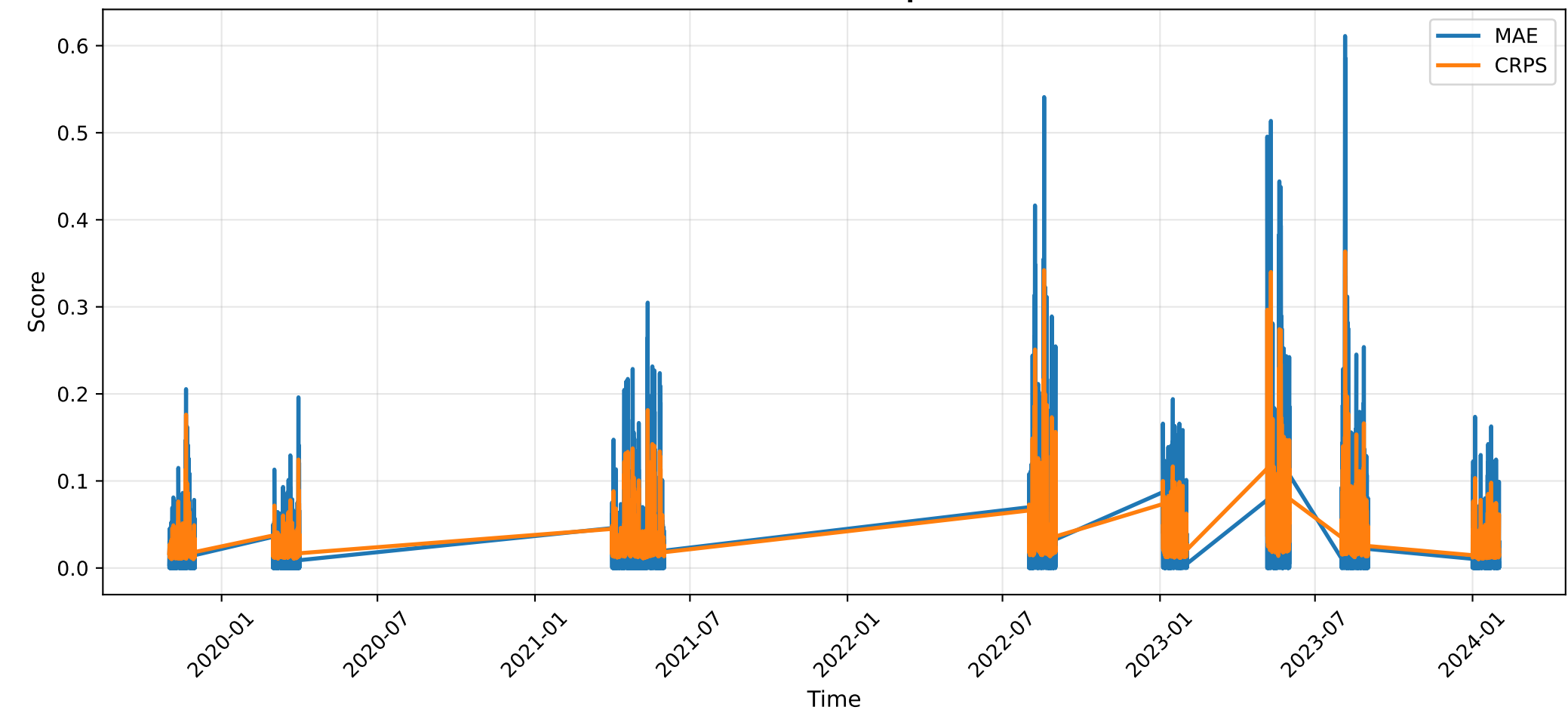


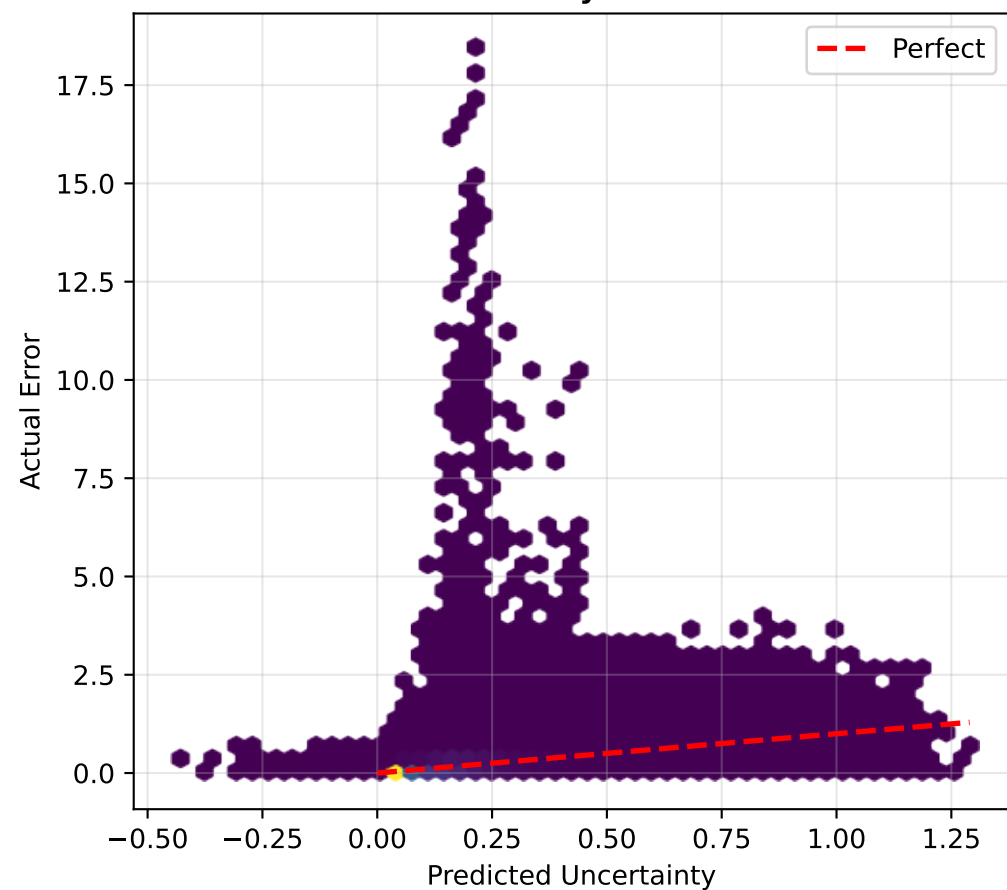
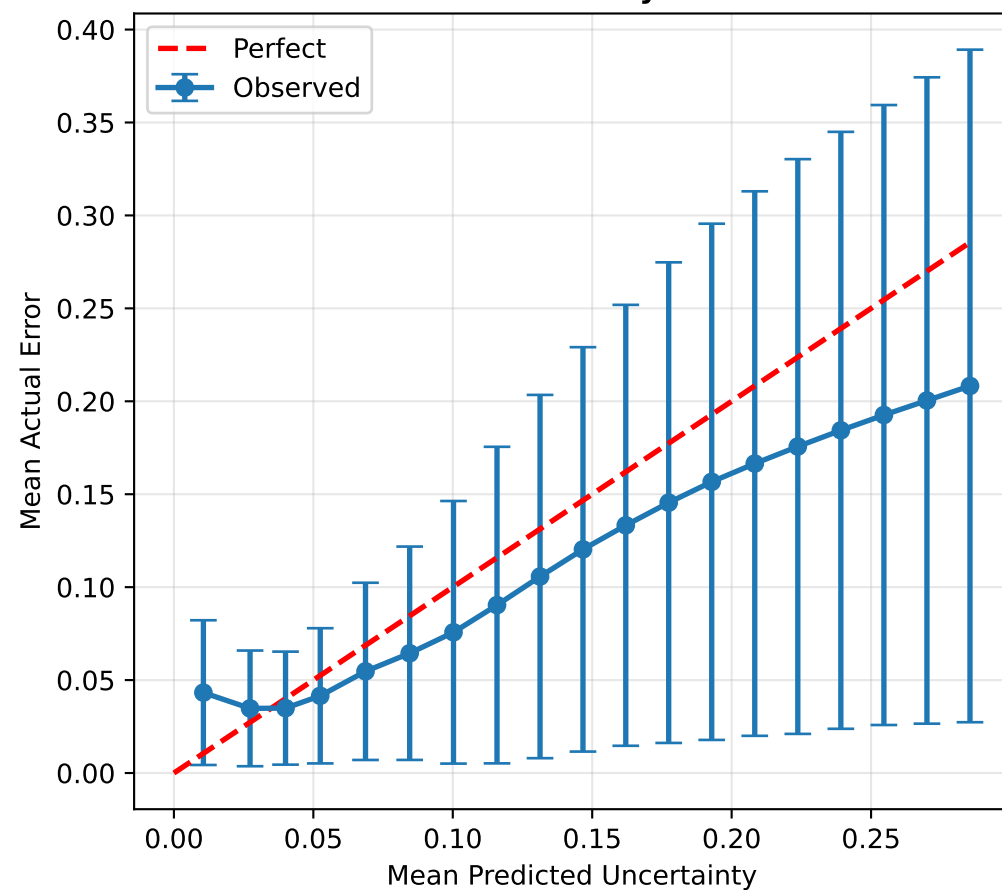
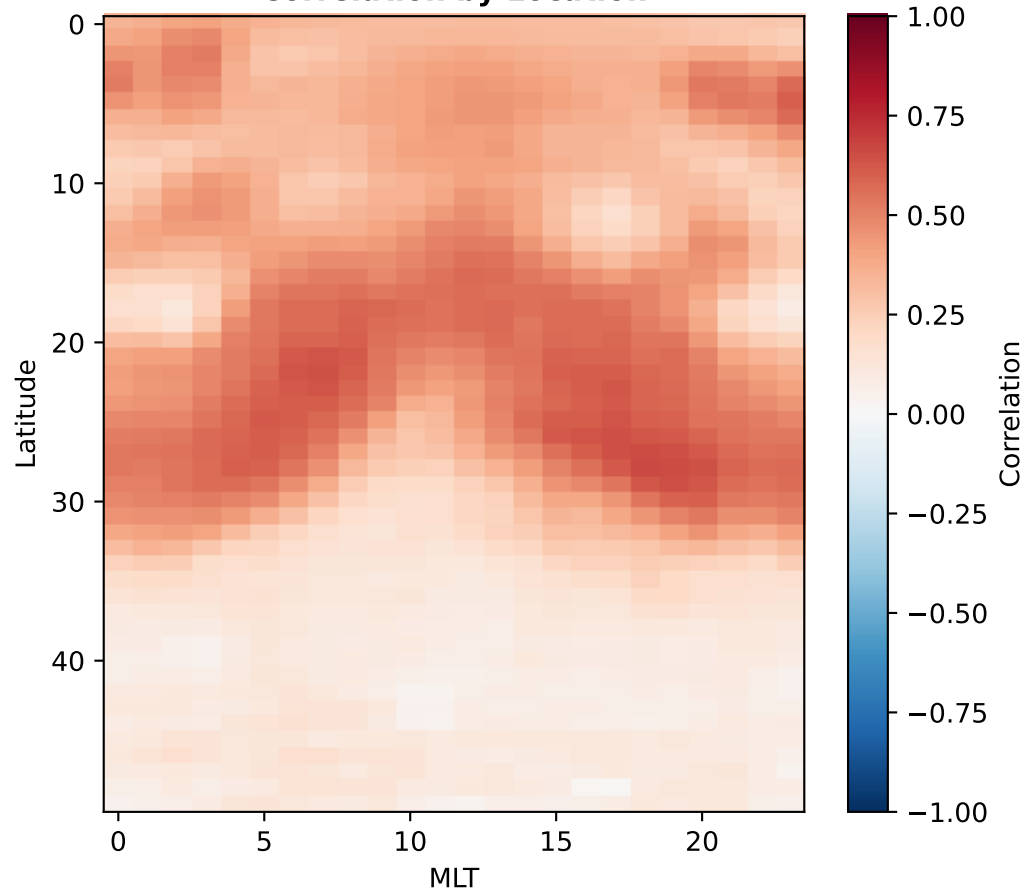
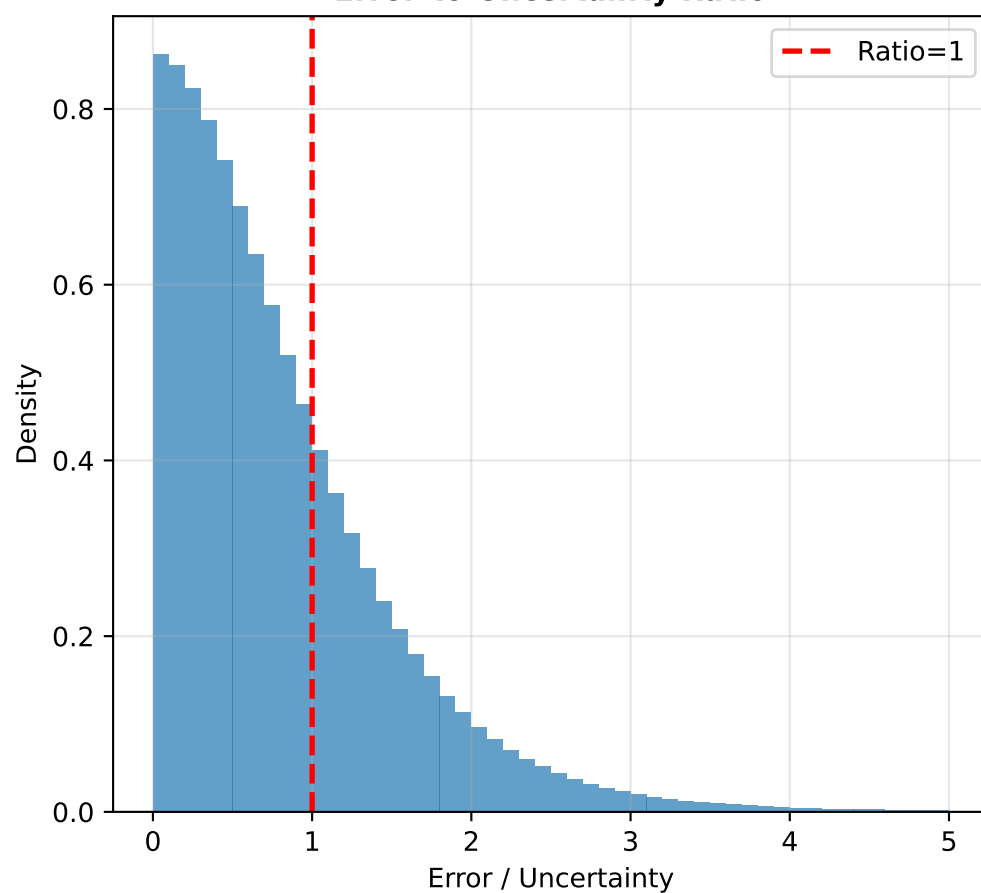
Calibration: Uncertainty vs Error**Z-Score Distribution****Calibration Curve****Q-Q Plot**

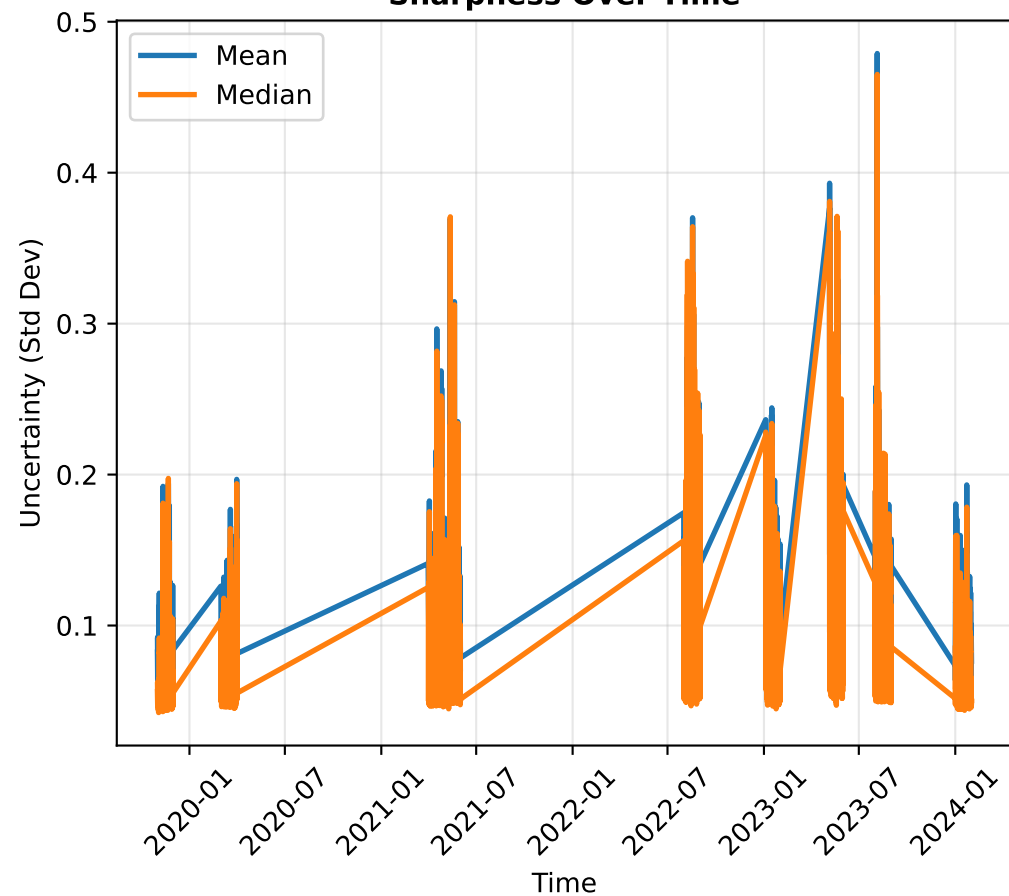
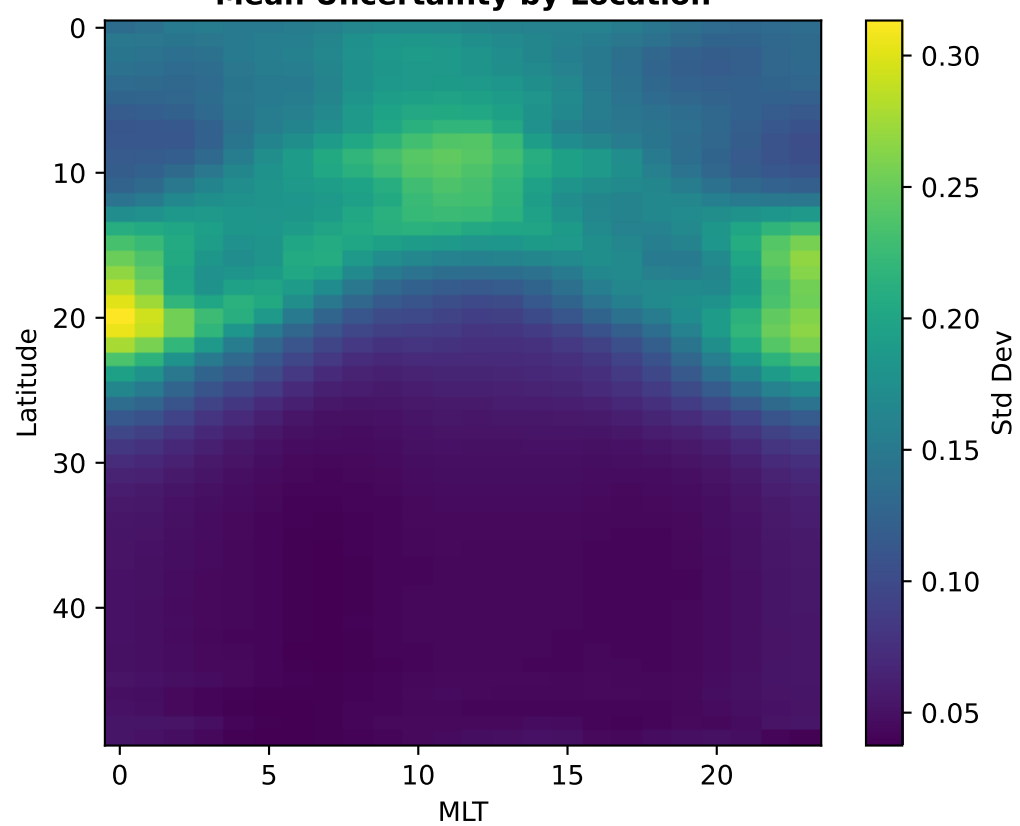
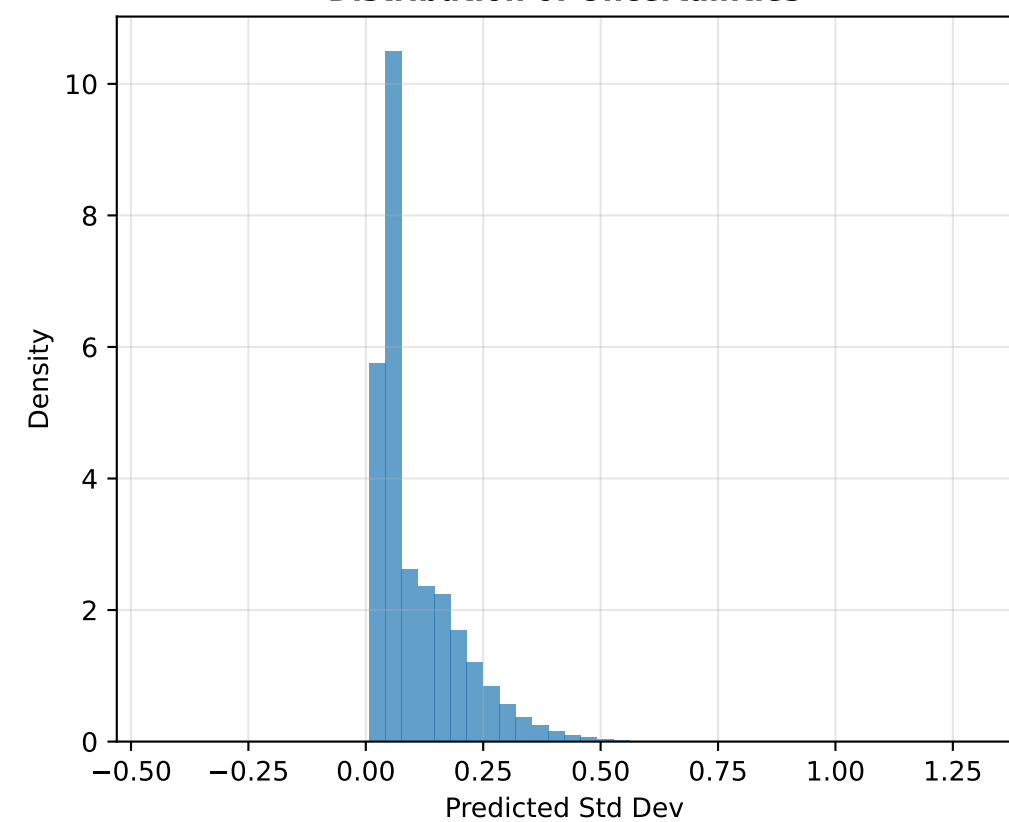
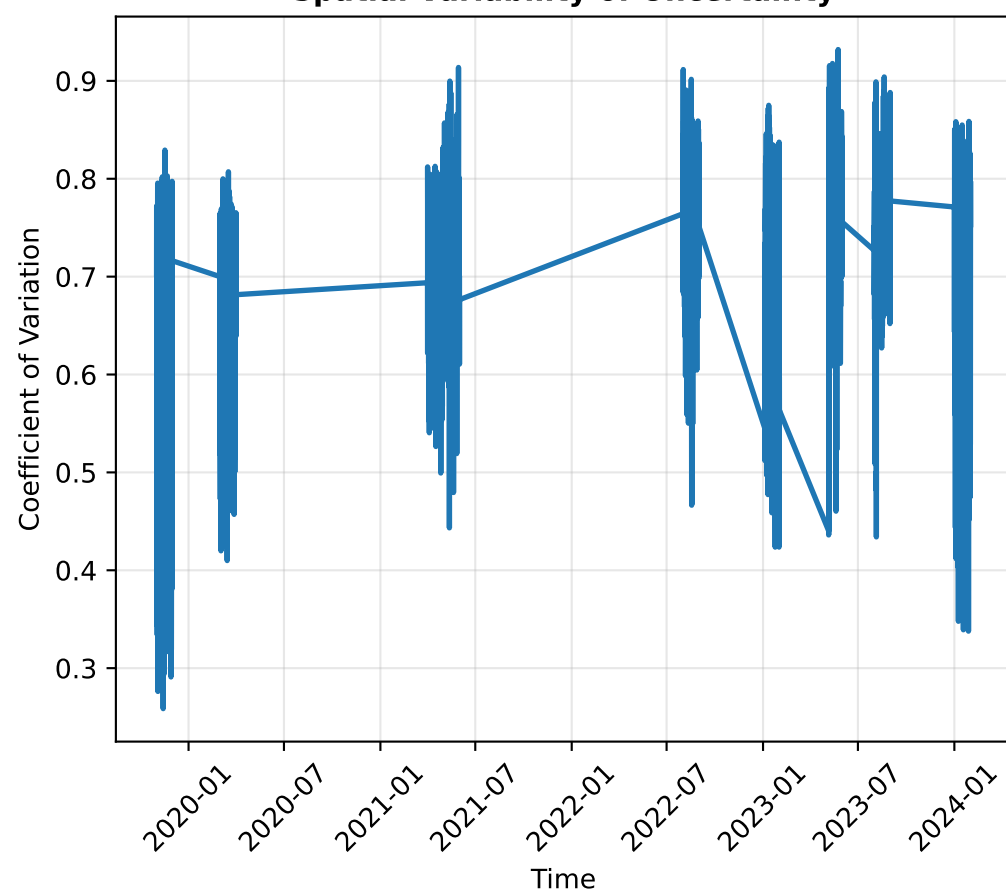
CRPS Over Time (Lat 65-75°, MLT 10-14)



MAE vs CRPS Comparison



Uncertainty vs Error**Binned Analysis****Correlation by Location****Error-to-Uncertainty Ratio**

Sharpness Over Time**Mean Uncertainty by Location****Distribution of Uncertainties****Spatial Variability of Uncertainty**

Numerical Metrics Summary

Deterministic Performance:

Point Correlation: 0.658

Point RMSE: 0.038

Point MAE: 0.024

Point Bias: -0.006

Uncertainty-Error Corr: 0.594

Optimal Lag: -2 timesteps

Max Cross-Correlation: 0.662

Uncertainty Quantification:

Coverage within 1σ : 99.7% (target: 68.3%)

Coverage within 2σ : 100.0% (target: 95.4%)

Z-score mean: 0.039 (target: 0)

Z-score std: 0.286 (target: 1)

Mean CRPS: 0.0282

CRPS/MAE Ratio: 1.155

Uncertainty-Error Correlation: 0.567

Median Error/Uncertainty Ratio: 0.639

Executive Summary

Point Performance:

Point prediction at Lat 65-75°, MLT 10-14 shows moderate correlation ($r=0.658$) with $RMSE=0.038$. Model exhibits negative bias of 0.006. Uncertainty-error correlation is 0.594, indicating strong awareness of prediction quality.

Spatial Performance:

Spatial analysis at MLT 10-14 shows mean $RMSE$ of 0.104 across all latitudes. Maximum errors (0.267) occur at high latitudes.

Temporal Alignment:

Model leads observations by 2 timesteps (anticipates changes). Maximum achievable correlation is 0.662.

Temporal Dynamics:

Autocorrelation analysis shows decorrelation timescales of 20 timesteps for observations and 20 timesteps for predictions. Model captures temporal persistence well.

Uncertainty Quality:

Uncertainty calibration is poor: 99.7% within 1σ (expected 68.3%), 100.0% within 2σ (expected 95.4%). Model is underconfident (uncertainties too wide). Z-score statistics: mean=0.039 (ideal: 0), std=0.286 (ideal: 1).

$CRPS/MAE = 1.16$; Uncertainty does not add value beyond point predictions. Mean $CRPS=0.0282$, Mean $MAE=0.0244$.

Uncertainty-error correlation is 0.567, indicating weak relationship. Median error/uncertainty ratio is 0.639 (underconfident (uncertainties too large)). Model shows moderate awareness of uncertainty patterns.

Uncertainty Characteristics:

Model sharpness: mean uncertainty=0.1073, median=0.0623. Temporal variability ($CV=0.365$) suggests condition-dependent uncertainty. Spatial variability ($CV=0.611$) indicates location-dependent confidence.